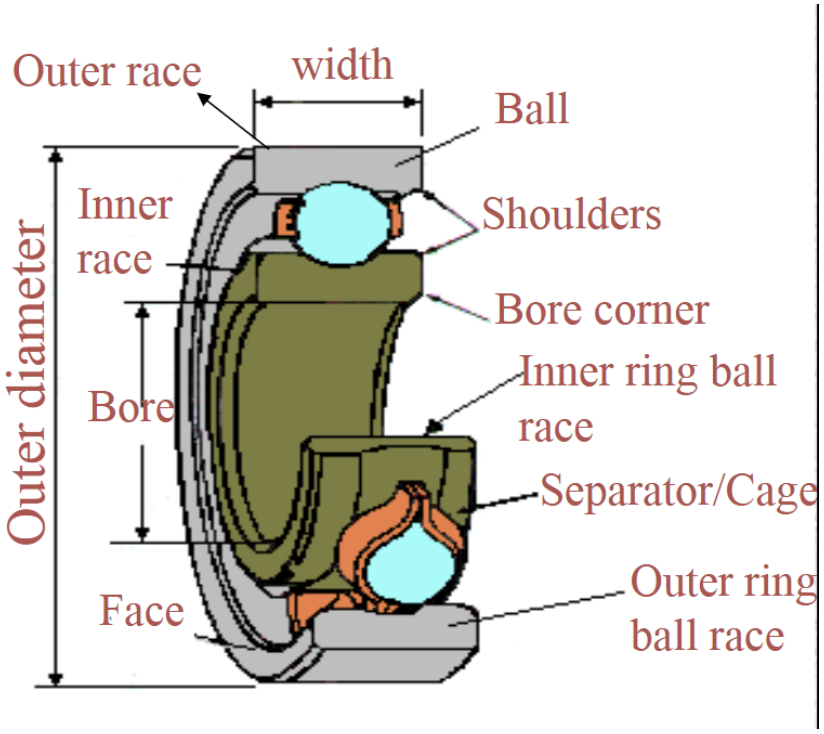
Bearing Selection

Bearings reduce friction between two members in motion.

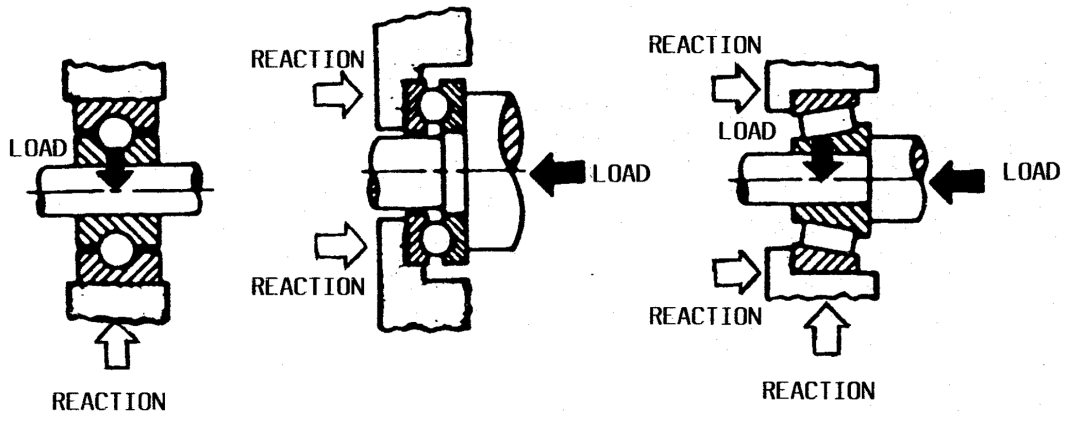
There are two major types of bearings:

- Plain Bearings are essentially sleeves that reduce friction, with some examples like
- Metallic Porous Impregnated Bearings
  - sintered metal sleeves like iron or bronze impregnated with a lubricant like oil or polymers
- Non-Metallic Plain Dry Bearings
  - sleeve made with polymers like PTFE (Teflon), nylon and carbon
- Hydrodynamic Fluid Film Bearings
  - the surfaces are kept away from each other either due to the rotating motion of one of the members or by external source. In the case of rotating members, the lubricant may be fed in under higher pressure.
- However, more typical bearings are rolling element bearings, or ball bearings.
- Ball bearings are able to run at higher speeds without the need of constant lubrication and cause less wear. However, they have a higher initial cost, require large space for the bearing to be mounted, do poorly under overloading or shock loading, are noisier under higher wear, and wear easier under foreign matter.

Shown are the parts of a typical ball bearing:



Bearings are designed to handle different kinds of loads, radial, axial, or both:



(a) Radial Load (b) Axial Load (c) Radial and Axial Loads

As for the types of roller element bearings, there are different roller elements and setups that may be used depending on loading conditions:

- Deep Groove Ball Bearings
  - most versatile and thus most common bearing
  - Supports mainly radial loads, but can support low axial loads
  - either has bearing code 6xxx or 16xxx

- Angular Contact Ball Bearings
  - for heavier axial loads to replace equivalent deep groove bearings
  - supports radial and single direction axial loads
  - for 30 deg contact angle bearing number 7xxx is used
  - for 40 deg contact angle, bearing 7xxxB is used
- Duplex Angular Contact Ball Bearings
  - two angular contact bearings clamped together to support axial loads in either direction
  - supports radial and axial loads
  - either in back to back arrangement: 7xxxBDB
  - or face to face arrangement: 7xxxBDF
- Double Row Self Aligning Ball Bearings
  - has alternating tracks with a spherical outer raceway to allow for misalignment
  - allows moderate radial and low axial loads
  - has bearing code 1xxx
- Thrust Ball Bearings
  - only allows axial loads
  - with 51xxx code for grooved raceways or
  - 9xxx/29xxx code for flat raceways.
- Double Row Spherical Ball Bearings
  - two rows of roller bearings with spherical outer raceway to allow for misalignment
  - supports radial and axial loads
  - has code 2xxx
- Roller Bearings
  - supports radial loads only
  - has code Nxxx or NUxxx

To determine what bearings to use, we refer to each bearing with the numbering system, in the form of XYZZ, where

X is the type code

Y is the series code

ZZ is the bore code

for the bore code, 00 to 03 are standard values:

00: 10mm bore

01: 12mm bore

02: 15mm bore

03: 17mm bore

but for 04 and upwards, the bore diameter is 5 times the bore code, so 04 has 4∗5 = 20mm bore

There are also other ISO terms, like

Rating Life (L_{10h} life)

the number of revolutions or hour of operations that 90% of a group of bearings can handle before fatigue develops.

Basic Dynamic Loading Rating (C_r)

the constant stationary radial load which a group of identical bearings of stationary outer rings can handle for a rating life of 1 mil revs of the inner ring

Dynamic Equivalent Bearing Load (P_r)

for a bearing undergoing axial and radial loads, Pr is the equivalent stationary radial load that has the same life

These terms allow us to form the equations:

$$f_h = f_n \times \frac{C_r}{P_r} \Rightarrow C_r = P_r \times \frac{f_h}{f_n}$$

and

$$P_r = XF_r + YF_a$$

however when $F_a = 0$, $X = 1, Y = 0$, thus

$$P_r = F_r$$

During bearing use of bearings larger than 12mm bore, either the inner or outer ring of the bearing must be secured properly to prevent the bearing from sliding along the axis

Inner Ring Clamping

Locknut & Lockwasher

A lockwasher engages into the slot of the shaft to clamp the inner ring, and the locknut locks the lockwasher in place, by bending the lockwasher's tabs into the slots of the locknut. The locknut is often chamfered to prevent forces acting on the outer ring.

Retaining Ring

A small ring, like a circlip, can be placed at the end of the bearing to limit the bearing from falling out, typically needing a groove in the shaft.

Adaptor Sleeves

mainly for self aligning bearings or spherical roller bearings, an adapter sleeve is used with a threaded portion to engage with a locknut and lockwasher, to clamp the bearing in place, giving easier mounting and dismounting due to it being a seperate subassembly, and machining on the shaft has higher tolerances.

External Ring Clamping

End Cover

An end cover clamps onto the outer ring by having clearance holes to allow bolts to clamp the end cover onto the housing, clamping the bearing in the process.

The end cover is fitted onto the shaft, hence a felt seal may be used to reduce friction of the rotating shaft.

Shaft & Housing Fit

To ensure smooth operation, the fit between the inner ring and shaft, as well as the fit between the outer ring and housing needs to be suitable for its application.

To ensure suitable fit, three points are to be considered:

Rotating rings of the bearings typically have an interference fit with the respective seating, except for location or single thrust bearings, where a push fit is used

Stationary rings of the bearings need not have a tight fit unless out of balance loads exceed the constant radial loading on the bearing (which causes slippage)

When bearings are fitted onto a housing or end cap, ensure that a shoulder is present and tall enough that the bearing can be located axially, and that the step between the should and the shaft should have a filet more than the bearing filet, or have a relief groove to prevent interference.